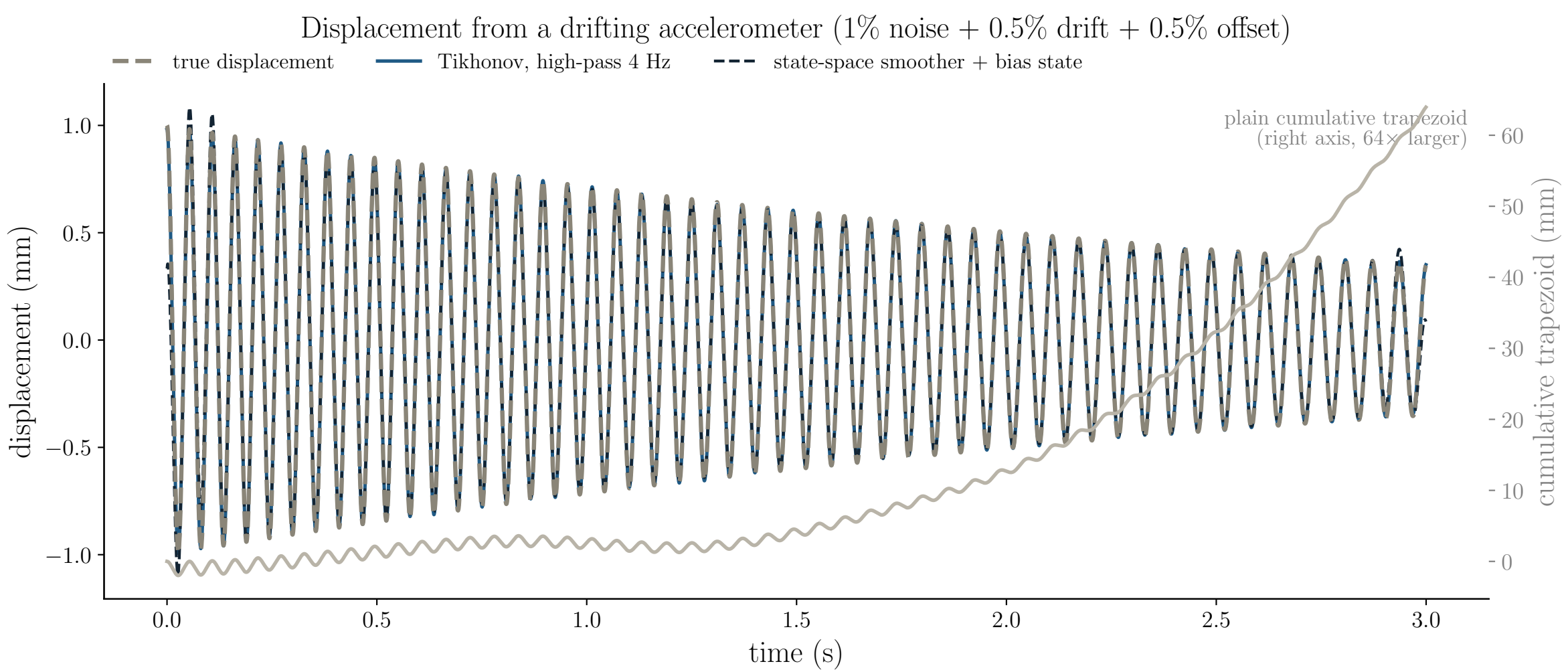

Displacement from a drifting accelerometer (1% noise + 0.5% drift + 0.5% offset)
true displacement
Tikhonov, high-pass 4 Hz
state-space smoother + bias state
plain cumulative trapezoid
(right axis, 64× larger)
displacement (mm)
cumulative trapezoid (mm)
time (s)
1.0
0.5
0.0
−0.5
−1.0
60
50
40
30
20
10
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0